HarvestIQ Ghana

Technical Report

Agricultural Decision Support via Machine Learning

Research Team

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Tamale, Ghana · 2025

1. Executive Summary

HarvestIQ Ghana is an end-to-end machine learning pipeline designed to support agricultural decision-making for smallholder farmers across Ghana. The project trains two separate regression models on 2,060 farm-level records spanning 10 Ghanaian regions and 8 crop types. The first model predicts crop yield in kilograms per hectare (Yield_kg_per_ha). The second predicts the number of days from planting to harvest (days_to_harvest). Both models use Random Forest as the primary algorithm, with Linear Regression as a performance baseline.

The full pipeline covers: raw data ingestion, duplicate removal, missing value imputation, feature engineering (including seasonality extraction), distribution analysis, outlier detection, categorical standardisation, correlation analysis, one-hot encoding, Lasso-based feature selection, model training, and evaluation via MAE, RMSE, and R^2.

The project is motivated by a real gap in Ghana's agricultural ecosystem: farmers make high-stakes planting decisions without access to data. HarvestIQ Ghana is a first step toward closing that gap using accessible ML tools and locally-grounded data.

2. Problem Statement

Smallholder farmers in Ghana operate with limited access to agronomic information. Decisions about which crop to plant, how much to expect from a harvest, and when that harvest will be ready are made primarily through experience and observation rather than data. This information gap has downstream effects on buyers who cannot accurately forecast supply, and on extension officers who lack evidence to tailor input recommendations by region and crop.

This project formalises three questions as machine learning objectives:

- **What should I plant?** — Crop type is a model input, not a target. Feature importance reveals which crops perform best under specific soil and rainfall conditions.
- **How much will I get?** — Modelled as regression on Yield_kg_per_ha.
- **When will it be ready?** — Modelled as regression on days_to_harvest.

3. Dataset

3.1 Overview

The dataset contains 2,060 farm-level records. After removing duplicates and filtering out implausible days_to_harvest values (negative values and values exceeding 400 days), approximately 2,000 clean records remain for modelling.

Attribute	Detail

Total Records (raw)	2,060
Records after cleaning	~2,000
Regions covered	10 Ghanaian regions
Crop types	8 (Cassava, Yam, Maize, Rice, Soybean, Groundnut, Cocoa, Plantain)
Missing values	~5% across key columns
Date range	2022–2023 planting seasons
Dataset type	Structured synthetic dataset (farm-level simulation)

3.2 Features

Column	Type	Role	Description
Farm_ID	Identifier	Dropped	Unique farm reference — no predictive signal
Crop_Type	Categorical	Feature	Type of crop planted
Farm_Location	Categorical	Feature	Ghanaian region
Nitrogen_Percent	Numeric	Feature	Soil nitrogen content (%)
Phosphorus_ppm	Numeric	Feature	Soil phosphorus (ppm)
Potassium_ppm	Numeric	Feature	Soil potassium (ppm)
Soil_pH	Numeric	Feature	Soil acidity / alkalinity
Rainfall_mm	Numeric	Feature	Seasonal rainfall — log-transformed
Planting_Date	Date	Engineered	Source for Planting_Month and Planting_Season
Harvest_Date	Date	Engineered	Used to compute days_to_harvest; then dropped
Yield_kg_per_ha	Numeric	Target 1	Crop yield in kilograms per hectare
days_to_harvest	Numeric	Target 2	Harvest_Date minus Planting_Date in days
Planting_Month	Numeric	Feature	Calendar month of planting (1–12)
Planting_Season	Categorical	Feature	Major (Mar–Aug) or Minor (Sep–Feb)

4. Pipeline Architecture

Step 1 — Data Inspection

Initial audit of shape, column types, and descriptive statistics. Identified 2,060 rows across 11 columns, with mixed data types and approximately 5% missing values in key columns.

Step 2 — Duplicate Removal

Duplicate rows were detected and removed before any imputation. Removing duplicates first prevents inflating the dataset with redundant observations that could bias imputed values.

Step 3 — Feature Engineering

Planting_Date and Harvest_Date were converted to datetime. days_to_harvest was computed as (Harvest_Date - Planting_Date).dt.days. Planting_Month and Planting_Season were extracted. Records with days_to_harvest ≤ 0 or > 400 were removed as data-entry errors.

Step 4 — Missing Value Imputation

Numeric columns were filled with their column mean. Categorical columns were filled with their column mode. Imputation was applied after feature engineering.

Step 5 — Categorical Data Cleaning

Farm_Location contained 17 name variants for 10 standard regions. A deterministic mapping dictionary standardised all variants to their canonical region names before encoding.

Step 6 — Distribution Analysis

Histograms with mean and median overlays were plotted for all numeric features. Skewness and kurtosis were computed. Rainfall_mm was log-transformed using numpy's log1p.

Step 7 — Outlier Detection

The IQR method was applied to all numeric columns. Outliers were identified and retained — in agricultural data, extreme values are often genuine observations, not errors.

Step 8 — Categorical Feature Exploration

Bar charts were generated for Crop_Type and Farm_Location to assess class balance.

Step 9 — Correlation Analysis

A lower-triangle correlation heatmap was generated. Soil nutrients showed moderate positive correlation with yield.

Step 10 — Encoding & Preprocessing

One-hot encoding applied to Crop_Type, Farm_Location, and Planting_Season (drop_first=True). Farm_ID and original date columns were dropped.

Step 11 — Lasso Feature Selection

All features standardised via StandardScaler. Lasso (alpha=0.01) fitted on Yield_kg_per_ha. Only features with non-zero coefficients were carried forward.

Step 12 — Model Training

Two separate regression pipelines trained: one for Yield_kg_per_ha and one for days_to_harvest. Each trains a Linear Regression baseline and a Random Forest (100 trees, random_state=42) on an 80/20 split.

Step 13 — Evaluation

MAE, RMSE, and R^2 computed for all four models. Actual vs. Predicted scatter plots and feature importance bar charts generated.

5. Modelling Approach

5.1 Why Two Separate Models

Yield and harvest duration are driven by fundamentally different signals. Yield is primarily a function of soil chemistry — nitrogen, phosphorus, potassium, and pH — and rainfall. Harvest duration is largely a function of crop biology: what was planted and when. Separate models allow each to specialise and be evaluated independently.

5.2 Algorithm Selection

Algorithm	Role	Rationale
Linear Regression	Baseline	Establishes a minimum performance floor. Fully interpretable. Fast to train. Assumes linear relati
Random Forest (100 trees)	Main Model	Captures non-linear relationships and feature interactions that linear models miss. Also provides

5.3 Feature Selection via Lasso

After one-hot encoding, the feature space expands substantially. Lasso regression applies L1 regularisation, which penalises weak features by shrinking their coefficients toward zero. This produces a lean feature set that improves generalisation and reduces the risk of overfitting.

5.4 Evaluation Metrics

Metric	What It Measures
MAE (Mean Absolute Error)	The average absolute difference between predicted and actual values, expressed in original units.
RMSE (Root Mean Squared Error)	Like MAE but penalises large errors more heavily due to squaring. More sensitive to outlier predictions.
R^2 (R-squared)	The proportion of variance in the target that the model explains. A score of 1.0 means perfect predictio

6. Results

The table below is ready to be populated after running all cells in the notebook. The evaluation function prints MAE, RMSE, and R^2 for each model at the end of Sections 13 and 14.

Model	Target Variable	Algorithm	MAE	RMSE	R^2
Baseline	Yield (kg/ha)	Linear Regression	—	—	—
Main	Yield (kg/ha)	Random Forest	—	—	—
Baseline	Days to Harvest	Linear Regression	—	—	—
Main	Days to Harvest	Random Forest	—	—	—

7. Key Technical Decisions

Log transformation on Rainfall_mm: Rainfall in Ghana is right-skewed. Without transformation, extreme values have disproportionate influence during training. Applying log1p compresses the upper tail, producing a more symmetric distribution.

Mean imputation for missing numeric values: Approximately 5% of values were missing. Mean imputation preserves the column's overall distribution. The known limitation is that imputed values carry no farm-level signal. KNN or model-based imputation would produce higher-quality fills.

Outliers retained, not removed: IQR detection flagged outliers in several numeric columns. In agricultural data, extreme values are often real — a particularly good harvest season, unusually high rainfall, or a very late planting. Removing them would discard genuine signal.

drop_first=True during one-hot encoding: Dropping the first dummy variable from each categorical feature avoids the dummy variable trap — a form of perfect multicollinearity. This is essential for Linear Regression and good practice for all models.

80/20 train-test split: A 20% holdout provides approximately 400 records for evaluation — sufficient to assess generalisation — while retaining 80% for training. A fixed random_state=42 ensures reproducibility.

Lasso alpha = 0.01: A small alpha applies mild regularisation, shrinking only genuinely weak features to zero while preserving features with moderate predictive signal. A larger alpha would risk eliminating useful predictors.

8. Limitations

Missing agronomic features: Crop yield is heavily influenced by fertiliser application, pest and disease pressure, irrigation status, and seed variety. None of these are present in the dataset. Their absence places a hard ceiling on predictive performance.

Single global model across eight crops: A single model encoding crop type as dummy variables must generalise across all eight crops. Separate per-crop models would achieve higher R^2 but require substantially more data per crop.

Small dataset: After cleaning, approximately 2,000 rows remain across 8 crops and 10 regions — roughly 250 records per crop. This is likely why Linear Regression remains competitive with Random Forest.

Mean imputation for missing target values: Approximately 5% of Yield_kg_per_ha values were imputed with column means. These values carry no real farm-level information and introduce noise into the yield model's training signal.

Synthetic dataset: The dataset was not sourced from verified real-world farm records. Patterns learned by the model may not transfer reliably to production use without validation against real MoFA Ghana records.

9. Recommended Next Steps

- Collect verified farm-level records from MoFA Ghana district offices or agricultural sensor networks
- Add fertiliser application rate and type, irrigation status, and seed variety as model features

- Build separate per-crop models once 500+ records per crop are available
- Replace mean imputation with KNN or model-based imputation for missing yield values
- Implement SHAP values for per-prediction explainability — critical for farmer-facing deployment
- Wrap both models in a Streamlit or mobile-friendly interface for field use by extension officers
- Extend the pipeline to HarvestIQ Post-Harvest: a crop quality grading classifier

10. Technology Stack

Library	Purpose
pandas	Data loading, cleaning, manipulation, and imputation
numpy	Numerical operations, log transformation
matplotlib / seaborn	Distribution histograms, boxplots, heatmaps, scatter plots, bar charts
scikit-learn	Lasso, StandardScaler, LinearRegression, RandomForestRegressor, metrics, train_test_split
scipy	Skewness and kurtosis computation for distribution analysis
Jupyter Notebook	Interactive development, cell-by-cell execution, inline visualisation